

Patrick Kough McFarlane

patmcfarla@gmail.com | (215) 499-1524 | Philadelphia, PA
[GitHub](#) | [Website](#) | [Twitter](#) | [LinkedIn](#) | [Google Scholar](#)

Education

Massachusetts Institute of Technology
Master of Science in Air Transportation

Cambridge, MA
Feb 2016

University of Notre Dame
Bachelor of Science in Aerospace Engineering

Notre Dame, IN
May 2014

Public Work

- **py_ball** — Python API wrapper for stats.nba.com with a focus on NBA and WNBA applications with 50k+ downloads
- **Evaluating NBA end-of-game decision-making**, *Journal of Sports Analytics*, 2019
- **He is Basketball's Undisputed MVP - of Wasting Time**, *The Wall Street Journal*

Skills

- **Methods:** Advanced Probability and Statistics, Data Mining, Machine Learning, Natural Language Processing, Optimization, Statistical Modeling, Stochastic Processes
- **Languages & Data:** Python, R, SQL, pandas, NumPy, scikit-learn, statsmodels, PyMC, Stan, Big Query
- **ML & Visualization:** PyTorch, TensorFlow, Keras, OpenCV, spaCy, NLTK, Plotly, Matplotlib, Agent AI
- **Engineering:** Flask, Dash, MLflow, PyCaret, Google Cloud Platform, Apache Airflow, Docker, GitHub Actions, dbt

Experience

Philadelphia Phillies

Director, Predictive Modeling

November 2023 – September 2026

Assistant Director, Baseball Research & Development

January 2022 – November 2023

- Directing the development and construction of the organization's forecasts of player performance by a team of ten data scientists.
- Integrating player performance forecasts into the club's decision-making processes to provide Baseball Operations with unbiased and data-driven guidance.
- Overseeing a technical recruitment process to ensure a sustainable, diverse hiring pipeline of data scientists for all levels of experience.
- Constructing a technical development program for data scientists to provide opportunities for advancement by identifying areas for improvement and growth.

Lead Quantitative Analyst – Player Evaluation

January 2020 – January 2022

- Leveraged player tracking, performance, and scouting data to build, maintain, and monitor predictive models to forecast future player performance for amateurs and professionals throughout the world.
- Consulted on potential player acquisitions for the entirety of the baseball calendar.
- Directed and managed the work of several analysts in support of player evaluation efforts.

Quantitative Analyst

January 2018 – January 2020

- Developed machine learning models to inform all aspects of baseball operations, including defensive positioning and defensive evaluation.
- Worked closely with stakeholders on implementation of research, models, and findings from the Baseball Research & Development department.

Bloomberg LP

Data Engineer

January 2017 – October 2017

The MITRE Corporation

Senior Systems Engineer

February 2016 – December 2016